

Multiple Choice (10 marks)

1. Which of the following is a biotic factor that could affect the growth rate of a population?
 - (a) Flash flood
 - (b) Tsunami
 - (c) Sudden reduction in the animal food resource
 - (d) Destruction of the ozone layer
 - (e) Drought
2. Which of the following statements about trypsin is NOT true?
 - (a) It is an organic compound made of proteins.
 - (b) It is a catalyst that alters the rate of a reaction.
 - (c) It is operative over a wide pH range.
 - (d) The rate of catalysis is affected by the concentration of substrate.
3. Certain nerve gases are known to cause a breakdown of cholinesterase. How would these gases affect the human body and why?
 - (a) Cause euphoria and hallucinations
 - (b) Improve memory and cognitive functions
 - (c) Increase physical strength and endurance
 - (d) Cause tremors, spasms, or even death
 - (e) Trigger temporary paralysis without pain
4. Why does milk "when kept in a refrigerator?"
 - (a) Enzymatic reactions with oxygen
 - (b) Lactose crystallization
 - (c) Loss of refrigeration power over time
 - (d) mesophiles
 - (e) Natural separation of milk components without bacterial influence
5. All of the following are examples of events that can prevent interspecific breeding EXCEPT
 - (a) the potential mates experience geographic isolation
 - (b) the potential mates experience behavioral isolation
 - (c) the potential mates have different courtship rituals
 - (d) the potential mates have similar breeding seasons

True / False (10 marks)

Answer True or False

6. True or False: The probability of drawing a face card first and an ace second, without replacement, is $1/110$.
7. True or False: Polytene chromosomes are produced only during meiosis and are not visible under a light microscope.
8. True or False: Animal cells have large vacuoles that store water, similar to plant cells.
9. True or False: Desert soils exhibit a high rate of leaching and nutrient cycling due to frequent rainfall and vegetation growth.
10. True or False: Sexual dimorphism is most often a result of parallel selection, where both sexes evolve traits simultaneously in response to similar pressures.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the genetic outcomes of mating a black male cat with a tortoiseshell female cat, focusing on sex-linked inheritance patterns and expected offspring phenotypes.
12. Discuss the key characteristics of arthropods that have contributed to their evolutionary success, analyzing the roles of their exoskeleton, organ systems, and nervous tissue organization.

End of Paper